Linear Algebra with Applications

Comprehensive Quiz

— All Learning Outcomes

Student Name: ______________________________

Date: ______________________________

Instructions

- Each section contains a **Worked Example** with a complete step-by-step solution, followed by a similar **Practice Question** to complete.
- Show all working clearly. Answers without supporting steps will not receive full credit.
- Calculators are permitted for numerical checks only.
- Unless stated otherwise, all entries are real numbers.

LO	Topic	Max Marks	Marks Obtained
1	Systems, Matrices & Determinants	24	________
2	Vector Spaces & Transformations	16	________
3	Orthogonality & Scalar Products	8	________
4	Eigenvalues & Eigenvectors	8	________
5	Numerical Methods & Canonical Forms	16	________
Total		**72**	________

Contents

1 Learning Outcome 1: Systems of Linear Equations, Matrix Arithmetic, and Determinants

1.1 AC 1.1 & 1.2 — Solving Linear Systems and Row Echelon Form

Assessment Criteria 1.1 & 1.2: Analyse and solve linear equations using matrices; evaluate row echelon form (REF) and reduced row echelon form (RREF) and their role in solving linear systems.

Problem. Solve the following system using Gaussian elimination. State the solution and verify it.

$$\begin{aligned} 2x_1 + 4x_2 - 2x_3 &= 2 \\ x_1 + 3x_2 + x_3 &= 7 \\ 3x_1 + 5x_2 - 3x_3 &= 3 \end{aligned}$$

Step 1. Write the augmented matrix $[A|\mathbf{b}]$:

$$\left[\begin{array}{ccc|c} 2 & 4 & -2 & 2 \\ 1 & 3 & 1 & 7 \\ 3 & 5 & -3 & 3 \end{array}\right]$$

Step 2. $R_1 \leftarrow \frac{1}{2}R_1$ (make the leading entry 1):

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 1 & 3 & 1 & 7 \\ 3 & 5 & -3 & 3 \end{array}\right]$$

Step 3. $R_2 \leftarrow R_2 - R_1$; $\quad R_3 \leftarrow R_3 - 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 6 \\ 0 & -1 & 0 & 0 \end{array}\right]$$

Step 4. $R_3 \leftarrow R_3 + R_2$ (eliminate below second pivot):

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 2 & 6 \end{array}\right] \quad \leftarrow \text{Row Echelon Form}$$

Step 5. $R_3 \leftarrow \frac{1}{2}R_3$; then back-substitute:

$$x_3 = 3 \qquad x_2 = 6-2(3) = 0 \qquad x_1 = 1-2(0)+(-1)(-3) = 1-0+3 \ \rightarrow\ x_1 = 1+3 = 4$$

Wait — re-check step 5: $x_1 = 1 - 2x_2 + x_3 = 1 - 2(0) + 3 = 4$.

Step 6. **Solution:** $(x_1, x_2, x_3) = (4,\ 0,\ 3)$.

Verification:

$$\begin{aligned} 2(4)+4(0)-2(3) &= 8-6=2 \checkmark \\ (4)+3(0)+(3) &= 7 \checkmark \\ 3(4)+5(0)-3(3) &= 12-9=3 \checkmark \end{aligned}$$

Note on REF: A matrix is in row echelon form when (i) all zero rows are at the bottom, (ii) each leading entry (pivot) is to the right of the pivot in the row above, and (iii) all entries below a pivot are zero. The REF in Step 4 exhibits all three properties.

Problem. **[LO 1]** *[AC 1.1, 1.2]* *(8 marks)*

Solve the following system by reducing the augmented matrix to row echelon form, then use back-substitution. Verify your solution.

$$\begin{aligned} 3x_1-6x_2+3x_3 &= 9 \\ 2x_1-x_2+5x_3 &= 7 \\ 5x_1-8x_2+3x_3 &= 11 \end{aligned}$$

(a) Write the augmented matrix and reduce it to row echelon form, clearly showing each elementary row operation. *(4)*

(b) State whether the system has a unique solution, infinitely many solutions, or no solution. Justify your answer from the REF. *(2)*

(c) Determine the solution (or solution set) and verify at least one equation. *(2)*

Answer Space

1.2 AC 1.3 — Matrix Notation and Operations

Assessment Criterion 1.3: Apply matrix notation and arithmetic operations, including addition, scalar multiplication, matrix multiplication, and the transpose.

Problem. Given

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \\ 2 & 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix},$$

compute (a) AB, (b) $A\mathbf{v}$, (c) B^TB.

Step 1. Compute AB (3×3 times $3 \times 2 = 3 \times 2$):

$$AB = \begin{bmatrix} 1(2) + (-2)(-1) + 3(4) & 1(1) + (-2)(3) + 3(0) \\ 0(2) + 4(-1) + (-1)(4) & 0(1) + 4(3) + (-1)(0) \\ 2(2) + 1(-1) + 0(4) & 2(1) + 1(3) + 0(0) \end{bmatrix} = \begin{bmatrix} 16 & -5 \\ -8 & 12 \\ 3 & 5 \end{bmatrix}$$

Step 2. Compute $A\mathbf{v}$ (3×3 times 3×1):

$$A\mathbf{v} = \begin{bmatrix} 1(1) + (-2)(-1) + 3(2) \\ 0(1) + 4(-1) + (-1)(2) \\ 2(1) + 1(-1) + 0(2) \end{bmatrix} = \begin{bmatrix} 1 + 2 + 6 \\ 0 - 4 - 2 \\ 2 - 1 \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 1 \end{bmatrix}$$

Step 3. Compute B^TB (2×3 times $3 \times 2 = 2 \times 2$):

$$B^T = \begin{bmatrix} 2 & -1 & 4 \\ 1 & 3 & 0 \end{bmatrix}, \qquad B^TB = \begin{bmatrix} 2(2) + (-1)(-1) + 4(4) & 2(1) + (-1)(3) + 4(0) \\ 1(2) + 3(-1) + 0(4) & 1(1) + 3(3) + 0(0) \end{bmatrix} = \begin{bmatrix} 21 & -1 \\ -1 & 10 \end{bmatrix}$$

Note that B^TB is **symmetric** (as expected for any B^TB).

Problem. **[LO 1]** *[AC 1.3]* *(6 marks)*

Given

$$P = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 2 \\ -1 & 4 & 1 \end{bmatrix}, \qquad Q = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ -1 & 3 \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix},$$

compute:

(a) PQ *(2)*

(b) $P\mathbf{u}$ *(2)*

(c) Q^TQ and comment on any special property of the result. *(2)*

Answer Space

1.3 AC 1.4 — Elementary Matrices and Partitioned Matrices

Assessment Criterion 1.4: Synthesise solutions involving elementary matrices and partitioned (block) matrices.

Problem.

(a) Write down the 3×3 elementary matrix E that multiplies row 2 by -3 and adds it to row 3.

(b) Express the row operation applied to $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix}$ as a matrix product EA, and compute EA.

(c) Find E^{-1} and verify $E^{-1}E = I$.

Step 1. Elementary matrix E**:** Start with I_3 and replace the $(3, 2)$ entry with -3 (add $-3\times$ row 2 to row 3):

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

Step 2. Compute EA**:**

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 4 + (-3)(1) & 0 + (-3)(3) & 1 + (-3)(2) \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 1 & -9 & -5 \end{bmatrix}$$

This is exactly the result of $R_3 \leftarrow R_3 - 3R_2$.

Step 3. Inverse E^{-1}**:** Reverse the operation: add $+3\times$ row 2 to row 3:

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

Verify:

$$E^{-1}E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 - 3 & -3 + 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \checkmark$$

Problem. **[LO 1]** *[AC 1.4]* *(6 marks)*

(a) Write the 3×3 elementary matrix F corresponding to swapping rows 1 and 3 (a row-interchange operation). *(1)*

(b) Let $B = \begin{bmatrix} 5 & 2 & 1 \\ -1 & 3 & 0 \\ 2 & -4 & 6 \end{bmatrix}$. Compute FB and describe the effect on B. *(2)*

(c) Find F^{-1}. What do you notice, and why does this make sense geometrically? *(2)*

(d) A matrix M is partitioned as $M = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$ where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 0 \end{bmatrix}$, $D = [2]$. Write out M in full and state its dimensions. *(1)*

Answer Space

1.4 AC 1.5 & 1.6 — Determinants and Applications

Assessment Criteria 1.5 & 1.6: Calculate determinants and assess their properties; analyse the role of determinants in solving linear systems (Cramer's Rule) and other applications.

Problem. Let $A = \begin{bmatrix} 3 & 1 & -2 \\ 0 & 4 & 1 \\ 2 & -1 & 3 \end{bmatrix}$.

(a) Calculate $\det(A)$ by cofactor expansion along the first column.
(b) State two properties of determinants demonstrated in your calculation.
(c) Use Cramer's Rule to solve $A\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$.

Step 1. Cofactor expansion along column 1:

$$\det(A) = 3 \cdot M_{11} - 0 \cdot M_{21} + 2 \cdot M_{31}$$

where M_{ij} is $(-1)^{i+j}$ times the (i, j) minor.

$$M_{11} = (+1) \det \begin{bmatrix} 4 & 1 \\ -1 & 3 \end{bmatrix} = (12 + 1) = 13$$
$$M_{21} = (-1) \det \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = (-1)(3 - 2) = -1 \quad \text{(not needed)}$$
$$M_{31} = (+1) \det \begin{bmatrix} 1 & -2 \\ 4 & 1 \end{bmatrix} = (1 + 8) = 9$$

$$\det(A) = 3(13) + 2(9) = 39 + 18 = \boxed{57}$$

Step 2. Properties illustrated:

- The row with a zero entry (R_2) contributes zero to the expansion, simplifying computation.
- $\det(A) \neq 0$, so A is **invertible** and the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.

Step 3. Cramer's Rule: $x_i = \dfrac{\det(A_i)}{\det(A)}$, where A_i replaces column i with $\mathbf{b}$.

$$A_1 = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 4 & 1 \\ 0 & -1 & 3 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & 1 & -2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 3 & 1 & 1 \\ 0 & 4 & 2 \\ 2 & -1 & 0 \end{bmatrix}$$

$$\det(A_1) = 1(12 + 1) - 1(6 - 0) + (-2)(-2 - 0) = 13 - 6 + 4 = 11$$
$$\det(A_2) = 3(6 - 0) - 1(0 - 2) + (-2)(0 - 4) = 18 + 2 + 8 = 28$$
$$\det(A_3) = 3(0 + 2) - 1(0 - 4) + 1(0 - 8) = 6 + 4 - 8 = 2$$

$$x_1 = \frac{11}{57}, \quad x_2 = \frac{28}{57}, \quad x_3 = \frac{2}{57}$$

Problem. **[LO 1]** *[AC 1.5, 1.6]* *(8 marks)*

Let $B = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \\ 4 & 0 & 1 \end{bmatrix}$.

(a) Calculate $\det(B)$ using cofactor expansion. Choose your row or column carefully to minimise computation. *(3)*

(b) Without further calculation, what does your answer tell you about the invertibility of B? *(1)*

(c) Use Cramer's Rule to solve the system $B\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Show the construction of each modified matrix B_i and the calculation of its determinant. *(4)*

Answer Space

2 Learning Outcome 2: Vector Spaces, Subspaces, and Linear Transformations

2.1 AC 2.1 & 2.2 — Vector Spaces, Linear Independence, Basis, and Dimension

Assessment Criteria 2.1 & 2.2: Define vector spaces and subspaces with examples; analyse linear independence, basis, and dimension.

Problem.

(a) Determine whether $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, where $\mathbf{v}_1 = (1, 2, -1)$, $\mathbf{v}_2 = (2, 0, 3)$, $\mathbf{v}_3 = (0, 4, -5)$.

(b) Find the dimension of the subspace $W = \text{Span}(S)$.

(c) Determine whether W is a subspace of $\mathbb{R}^3$ and state two subspace axioms it satisfies.

Step 1. Test for linear independence: set $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$ and row-reduce:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 4 \\ -1 & 3 & -5 \end{bmatrix} \xrightarrow{R_2 - 2R_1,\ R_3 + R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -4 & 4 \\ 0 & 5 & -5 \end{bmatrix} \xrightarrow{R_3 + \frac{5}{4}R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -4 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

The system has a free variable (c_3). Setting $c_3 = 1$: $c_2 = 1$, $c_1 = -2$. Hence $-2\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, so S is **linearly dependent**.

Verify: $-2(1, 2, -1) + (2, 0, 3) + (0, 4, -5) = (-2 + 2 + 0,\ -4 + 0 + 4,\ 2 + 3 - 5) = (0, 0, 0)$ ✓

Step 2. Dimension of W: The row-reduced matrix has **2 non-zero rows** (2 pivots), so $\dim(W) = 2$. A basis for W is $\{\mathbf{v}_1, \mathbf{v}_2\}$ (the pivot columns).

Step 3. W is a subspace of $\mathbb{R}^3$ because it is defined as a span:

- **Closure under addition:** if $\mathbf{u}, \mathbf{w} \in W$ then $\mathbf{u} = \sum \alpha_i \mathbf{v}_i$ and $\mathbf{w} = \sum \beta_i \mathbf{v}_i$, so $\mathbf{u} + \mathbf{w} = \sum(\alpha_i + \beta_i)\mathbf{v}_i \in W$.
- **Closure under scalar multiplication:** $\lambda\mathbf{u} = \sum(\lambda\alpha_i)\mathbf{v}_i \in W$.

Problem. **[LO 2]** *[AC 2.1, 2.2]* *(8 marks)*

Let $S = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$ where $\mathbf{w}_1 = (1, 0, 2, 1)$, $\mathbf{w}_2 = (0, 1, -1, 2)$, $\mathbf{w}_3 = (2, 1, 3, 4)$, $\mathbf{w}_4 = (1, -1, 3, -1)$.

(a) Row-reduce the matrix whose rows are $\mathbf{w}_1, \ldots, \mathbf{w}_4$. Show each step. *(3)*

(b) Determine whether S is linearly independent or dependent. If dependent, express one vector as a linear combination of the others. *(2)*

(c) State the dimension of $W = \text{Span}(S)$ and give a basis for W. *(2)*

(d) Verify that $\mathbf{0} \in W$, and use this to confirm W is a subspace of $\mathbb{R}^4$. *(1)*

Answer Space

2.2 AC 2.3 & 2.4 — Linear Transformations and Matrix Representations

Assessment Criteria 2.3 & 2.4: Define and apply linear transformations; analyse their matrix representations and the concept of similarity.

Problem. Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 - x_2 + x_3 \\ x_1 + 3x_3 \end{pmatrix}$.

(a) Show that T is a linear transformation.
(b) Find the standard matrix $[T]$ of T.
(c) Find $\ker(T)$ and $\operatorname{range}(T)$, and verify the Rank-Nullity Theorem.

Step 1. Linearity: For T to be linear, $T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$.

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \begin{bmatrix} 2(\alpha u_1 + \beta v_1) - (\alpha u_2 + \beta v_2) + (\alpha u_3 + \beta v_3) \\ (\alpha u_1 + \beta v_1) + 3(\alpha u_3 + \beta v_3) \end{bmatrix} = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}) \checkmark$$

(Each component is a linear combination of the inputs — the transformation is linear.)

Step 2. Standard matrix: Apply T to each standard basis vector $\mathbf{e}_i$:

$$T(\mathbf{e}_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_3) = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \Rightarrow [T] = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

Step 3. Kernel (null space): Solve $[T]\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -5 \end{bmatrix} \Rightarrow x_2 = 5x_3,\ x_1 = -3x_3$$

$\ker(T) = \operatorname{Span}\left\{ \begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} \right\}$, so $\operatorname{nullity}(T) = 1$.

Range: Two pivot columns $\Rightarrow \operatorname{rank}(T) = 2$; range $= \mathbb{R}^2$.

Rank-Nullity: $\text{rank} + \text{nullity} = 2 + 1 = 3 = \dim(\mathbb{R}^3)$ ✓

Problem. **[LO 2]** *[AC 2.3, 2.4]* *(8 marks)*

Define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 \\ 3x_1 - x_3 \\ x_2 + 4x_3 \end{pmatrix}$.

(a) Find the standard matrix $[T]$. *(2)*
(b) Find the kernel and range of T. State $\operatorname{rank}(T)$ and $\operatorname{nullity}(T)$ and verify the Rank-Nullity Theorem. *(4)*
(c) Two matrices A and B are **similar** if there exists an invertible matrix P such

that $B = P^{-1}AP$. Given $[T]$ above and $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, compute $P^{-1}[T]P$.

What does similarity preserve? *(2)*

Answer Space

3 Learning Outcome 3: Orthogonality and Scalar Products

3.1 AC 3.1 & 3.2 — Scalar Products, Orthogonality, and Least Squares

Assessment Criteria 3.1 & 3.2: Analyse scalar products and orthogonality in $\mathbb{R}^n$; apply orthogonality to least squares problems and construct orthonormal sets.

Problem. Let $\mathbf{u}_1 = (1,1,0)$, $\mathbf{u}_2 = (1,-1,1)$, $\mathbf{b} = (3,1,2)$.

(a) Show that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set, then normalise to get an orthonormal set.
(b) Project $\mathbf{b}$ onto $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$.
(c) Find the component of $\mathbf{b}$ orthogonal to W and verify your decomposition.

Step 1. Orthogonality check: $\mathbf{u}_1 \cdot \mathbf{u}_2 = (1)(1) + (1)(-1) + (0)(1) = 0$ ✓

Normalise: $\|\mathbf{u}_1\| = \sqrt{1+1+0} = \sqrt{2}$, $\|\mathbf{u}_2\| = \sqrt{1+1+1} = \sqrt{3}$

$$\hat{\mathbf{u}}_1 = \tfrac{1}{\sqrt{2}}(1,1,0), \quad \hat{\mathbf{u}}_2 = \tfrac{1}{\sqrt{3}}(1,-1,1)$$

Orthonormal set: $\{\hat{\mathbf{u}}_1, \hat{\mathbf{u}}_2\}$.

Step 2. Orthogonal projection of b onto W:

$$\hat{\mathbf{b}} = \frac{\mathbf{b}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{b}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\mathbf{u}_2$$

$$\mathbf{b}\cdot\mathbf{u}_1 = 3+1+0 = 4, \quad \mathbf{u}_1\cdot\mathbf{u}_1 = 2 \Rightarrow \text{coeff}_1 = 2$$

$$\mathbf{b}\cdot\mathbf{u}_2 = 3-1+2 = 4, \quad \mathbf{u}_2\cdot\mathbf{u}_2 = 3 \Rightarrow \text{coeff}_2 = \tfrac{4}{3}$$

$$\hat{\mathbf{b}} = 2(1,1,0) + \tfrac{4}{3}(1,-1,1) = (2,2,0) + (\tfrac{4}{3}, -\tfrac{4}{3}, \tfrac{4}{3}) = \left(\tfrac{10}{3},\ \tfrac{2}{3},\ \tfrac{4}{3}\right)$$

Step 3. Orthogonal component: $\mathbf{b} - \hat{\mathbf{b}} = (3 - \tfrac{10}{3},\ 1 - \tfrac{2}{3},\ 2 - \tfrac{4}{3}) = (-\tfrac{1}{3},\ \tfrac{1}{3},\ \tfrac{2}{3})$.

Verify: $(-\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{2}{3})\cdot(1,1,0) = -\tfrac{1}{3}+\tfrac{1}{3} = 0$ ✓; $(-\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{2}{3})\cdot(1,-1,1) = -\tfrac{1}{3}-\tfrac{1}{3}+\tfrac{2}{3} = 0$ ✓

Problem. [LO 3] *[AC 3.1, 3.2]* *(8 marks)*

Let $\mathbf{v}_1 = (2,1,-1)$, $\mathbf{v}_2 = (1,-1,0)$, and $\mathbf{c} = (4,2,1)$.

(a) Verify that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal set, then construct an orthonormal set $\{\hat{\mathbf{v}}_1, \hat{\mathbf{v}}_2\}$. *(2)*
(b) Compute the orthogonal projection $\hat{\mathbf{c}}$ of $\mathbf{c}$ onto $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. *(3)*
(c) Find the component $\mathbf{c} - \hat{\mathbf{c}}$ orthogonal to W and verify it is perpendicular to both $\mathbf{v}_1$ and $\mathbf{v}_2$. *(2)*
(d) Interpret the decomposition $\mathbf{c} = \hat{\mathbf{c}} + (\mathbf{c} - \hat{\mathbf{c}})$ in the context of a least-squares problem: if $A = \begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2\end{bmatrix}$ (columns), what does $\hat{\mathbf{c}}$ represent? *(1)*

Answer Space

4 Learning Outcome 4: Eigenvalues and Eigenvectors

4.1 AC 4.1 & 4.2 — Eigenanalysis and Applications

Assessment Criteria 4.1 & 4.2: Analyse eigenvalues and eigenvectors including complex eigenvalues; apply eigenvalues to solving linear systems and understanding transformations.

Problem. Let $A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$.

(a) Find the eigenvalues of A.
(b) Find an eigenvector for each eigenvalue.
(c) Diagonalise A: express $A = PDP^{-1}$ where D is diagonal.
(d) Use the diagonalisation to compute A^3.

Step 1. Characteristic equation: $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 4-\lambda & -2 \\ 1 & 1-\lambda \end{bmatrix} = (4-\lambda)(1-\lambda) + 2 = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3) = 0$$

Eigenvalues: $\lambda_1 = 2$, $\lambda_2 = 3$.

Step 2. Eigenvectors:

For $\lambda_1 = 2$: $(A - 2I)\mathbf{x} = \mathbf{0}$: $\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \mathbf{x} = \mathbf{0} \Rightarrow x_1 = x_2$; eigenvector $\mathbf{p}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

For $\lambda_2 = 3$: $(A - 3I)\mathbf{x} = \mathbf{0}$: $\begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \mathbf{x} = \mathbf{0} \Rightarrow x_1 = 2x_2$; eigenvector $\mathbf{p}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Step 3. Diagonalisation:

$$P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad P^{-1} = \frac{1}{\det P} \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

Verify: $AP = PD$ iff $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$: $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ✓; $A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ ✓

Step 4. $A^3 = PD^3P^{-1}$**:**

$$D^3 = \begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 8 & 54 \\ 8 & 27 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 46 & -38 \\ 19 & -11 \end{bmatrix}$$

Problem. [**LO 4**] *[AC 4.1, 4.2]* *(8 marks)*

Let $B = \begin{bmatrix} 3 & -5 \\ 1 & -1 \end{bmatrix}$.

(a) Find the characteristic polynomial of B and compute its eigenvalues. Note: the discriminant is negative — state the complex eigenvalues in the form $a \pm bi$. *(2)*

(b) For each eigenvalue, find a corresponding (complex) eigenvector. *(2)*

(c) Describe geometrically what a transformation with complex eigenvalues $a \pm bi$ does to vectors in $\mathbb{R}^2$. (Hint: consider the modulus $r = \sqrt{a^2 + b^2}$ and argument $\theta = \arctan(b/a)$.) *(2)*

(d) Find $\operatorname{tr}(B)$ and $\det(B)$ and relate them to the sum and product of the eigenvalues respectively. *(2)*

Answer Space

5 Learning Outcome 5: Numerical Methods and Canonical Forms

5.1 AC 5.1 & 5.2 — Floating-Point Numbers and Numerical Gaussian Elimination

Assessment Criteria 5.1 & 5.2: Analyse floating-point representation; apply Gaussian elimination with pivoting strategies and matrix norms.

Problem.

(a) Represent $x = 0.1$ in IEEE 754 single-precision floating-point (32-bit). Why can it not be stored exactly?

(b) Use Gaussian elimination **with partial pivoting** to solve: $\begin{bmatrix} 0.001 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. Contrast with naive elimination (no pivoting) to show the benefit.

Step 1. Floating-point representation of 0.1**:** $0.1 = 0.0\overline{0011}_2$ (the binary expansion is periodic/non-terminating). With 23 explicit mantissa bits, the stored value is approximately $0.10000000149\ldots \neq 0.1$, introducing round-off error. IEEE 754 single: sign 0, exponent $-4 + 127 = 123 = (01111011)_2$, mantissa truncated to 23 bits.

Step 2. Naive elimination (without pivoting): Multiplier $m = 1/0.001 = 1000$. After elimination: $\begin{bmatrix} 0.001 & 1 \\ 0 & 2 - 1000 \end{bmatrix} = \begin{bmatrix} 0.001 & 1 \\ 0 & -998 \end{bmatrix} \begin{bmatrix} & | \, 1 \\ | & -997 \end{bmatrix}$

$x_2 = -997/-998 \approx 0.999$; $x_1 = (1 - 1 \cdot 0.999)/0.001 \approx 1.0$. This is numerically unstable if 0.001 suffers catastrophic cancellation.

Step 3. With partial pivoting: Swap rows (pivot on largest in column 1):

$$\begin{bmatrix} 1 & 2 \\ 0.001 & 1 \end{bmatrix} \begin{bmatrix} | \\ | \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \xrightarrow{R_2 - 0.001 R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0.998 \end{bmatrix} \begin{bmatrix} | \\ | \end{bmatrix} \begin{bmatrix} 3 \\ 0.997 \end{bmatrix}$$

$x_2 = 0.997/0.998 \approx 0.999$; $x_1 = 3 - 2(0.999) = 1.002$.

Exact solution: A^{-1} gives $x_1 \approx 1.0$, $x_2 \approx 1.0$. Partial pivoting gives a more accurate result by avoiding division by a near-zero pivot.

∞-norm of A: $\|A\|_\infty = \max(\text{row sums}) = \max(|0.001| + |1|,\ |1| + |2|) = 3$.

Problem. [LO 5] *[AC 5.1, 5.2]* *(8 marks)*

(a) A floating-point system uses 4-bit mantissa (including sign) and 3-bit exponent in base 2. Represent $x = 5.5$ in this system and identify the machine epsilon $\varepsilon_{\text{mach}}$. What is the smallest positive representable number in this system? *(3)*

(b) Apply Gaussian elimination **with partial pivoting** to solve: $\begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & 2 \\ -2 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 22 \\ -2 \end{bmatrix}$. Show each pivot selection and row swap. Verify your solution. *(5)*

Answer Space

5.2 AC 5.3 & 5.4 — Jordan Canonical Form and Nilpotent Operators

Assessment Criteria 5.3 & 5.4: Analyse nilpotent operators and the Jordan canonical form; apply canonical form concepts in context.

Problem. Let $N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$.

(a) Show that N is nilpotent and find its index of nilpotency.
(b) Write A in Jordan canonical form. Identify the Jordan block(s).
(c) Compute $e^A = e^{3I+N}$ using the matrix exponential series. (Hint: $e^{3I+N} = e^{3I}e^N$.)

Step 1. Nilpotency of N:

$$N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad N^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

$N^2 \neq O$ but $N^3 = O$, so N is nilpotent with **index** $k = 3$.

Step 2. Jordan form of A: The only eigenvalue of A is $\lambda = 3$ (multiplicity 3). Since $A - 3I = N \neq 0$ and $(A-3I)^2 = N^2 \neq 0$ but $(A-3I)^3 = 0$, the Jordan block has size 3. The Jordan canonical form is:

$$J = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

(Here A is already in Jordan form; the single 3×3 block corresponds to eigenvalue $\lambda = 3$.)

Step 3. Matrix exponential e^A: Since $3I$ and N commute, $e^{3I+N} = e^{3I}e^N$.

$$e^{3I} = e^3 I, \qquad e^N = I + N + \tfrac{1}{2!}N^2 = \begin{bmatrix} 1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$e^A = e^3 \begin{bmatrix} 1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Significance: The Jordan form reveals that A has a defective eigenvalue (geometric multiplicity $1 <$ algebraic multiplicity 3), meaning A is *not* diagonalisable over $\mathbb{R}$.

Problem. [**LO 5**] *[AC 5.3, 5.4]* *(8 marks)*

Let $C = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$.

(a) Identify the eigenvalues of C and their algebraic and geometric multiplicities. *(2)*

(b) Write the Jordan canonical form J of C. Draw the Jordan block structure, labelling each block with its eigenvalue and size. *(3)*

(c) Let $M = C - 2I$. Show that M is nilpotent and find its index of nilpotency. *(2)*

(d) State one practical application of the Jordan form (e.g. in differential equations or matrix powers). *(1)*

Answer Space

Formula Sheet

Key Formulae — for Reference

Determinants

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc, \qquad \det(A) = \sum_{j=1}^{n} a_{ij}(-1)^{i+j} M_{ij} \text{ (cofactor expansion along row } i\text{)}$$

Cramer's Rule $x_i = \dfrac{\det(A_i)}{\det(A)}$, where A_i replaces column i of A with $\mathbf{b}$.

Inverse of 2×2 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Orthogonal Projection $\mathrm{proj}_W \mathbf{b} = \displaystyle\sum_{i=1}^{k} \frac{\mathbf{b} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i} \mathbf{u}_i$ (where $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an orthogonal basis for W)

Eigenvalues $\det(A - \lambda I) = 0$; eigenvectors satisfy $(A - \lambda I)\mathbf{x} = \mathbf{0}$.

Trace–Eigenvalue relations $\mathrm{tr}(A) = \sum \lambda_i$, $\det(A) = \prod \lambda_i$

Diagonalisation $A = PDP^{-1}$, $A^k = PD^kP^{-1}$

Matrix Exponential $e^A = \displaystyle\sum_{k=0}^{\infty} \frac{A^k}{k!}$; if $A = PJP^{-1}$, then $e^A = Pe^JP^{-1}$

Rank-Nullity Theorem $\mathrm{rank}(T) + \mathrm{nullity}(T) = \dim(\text{domain})$

Matrix Norms $\|A\|_\infty = \max_i \sum_j |a_{ij}|$, $\|A\|_1 = \max_j \sum_i |a_{ij}|$

Nilpotent operator $N^k = O$ for some positive integer k (index of nilpotency = smallest such k)

Jordan Block of size m for eigenvalue λ:

$$J_m(\lambda) = \begin{bmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix}_{m \times m}$$